

◇ Part 5 — Model Training

1. What is the slope?

The slope (coefficients) shows how much the output (Marks) changes when input features like Study_Hours, Attendance, and Assignments change. Each feature has its own impact on prediction.

2. What is the intercept?

The intercept is the base value of the prediction when all input features are zero. It represents starting value of the model.

3. What do slope and intercept mean?

Together, slope and intercept form the Linear Regression equation. This equation is used by the model to predict marks based on input features.

Part 7 — Comparison

4. Which student had the highest prediction error?

The student whose difference between actual marks and predicted marks is highest has the maximum error. This can be found using the Difference column in the comparison table.

◇ Part 8 — Evaluation

5. What is MAE?

Mean Absolute Error (MAE) is the average difference between actual and predicted values. It shows how much error the model is making. Lower MAE means better model performance.

6. What is R² Score?

R² Score shows how well the model fits the data. It ranges from 0 to 1. A value closer to 1 means the model is performing very well.

7. Is your model performing well?

Yes, if the R² score is close to 1 and MAE is low, then the model is performing well and giving accurate predictions.

◇ Part 10 — Research Questions

8. What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence where machines learn patterns from data and make predictions without being explicitly programmed. It improves automatically using experience and data.

9. What is Linear Regression?

Linear Regression is a supervised learning algorithm used to predict continuous values. It finds a linear relationship between input features and output labels.

10. What are Features?

Features are input variables used by the model to make predictions. Example: Study_Hours, Attendance, Assignments.

11. What are Labels?

Labels are output variables that the model predicts. Example: Marks.

12. Difference between Training and Testing Data?

Training data is used to teach the model. Testing data is used to evaluate model performance on unseen data.

13. Real-life applications of Machine Learning?

Machine Learning is used in recommendation systems, fraud detection, healthcare diagnosis, self-driving cars, voice assistants, and many real-world applications.